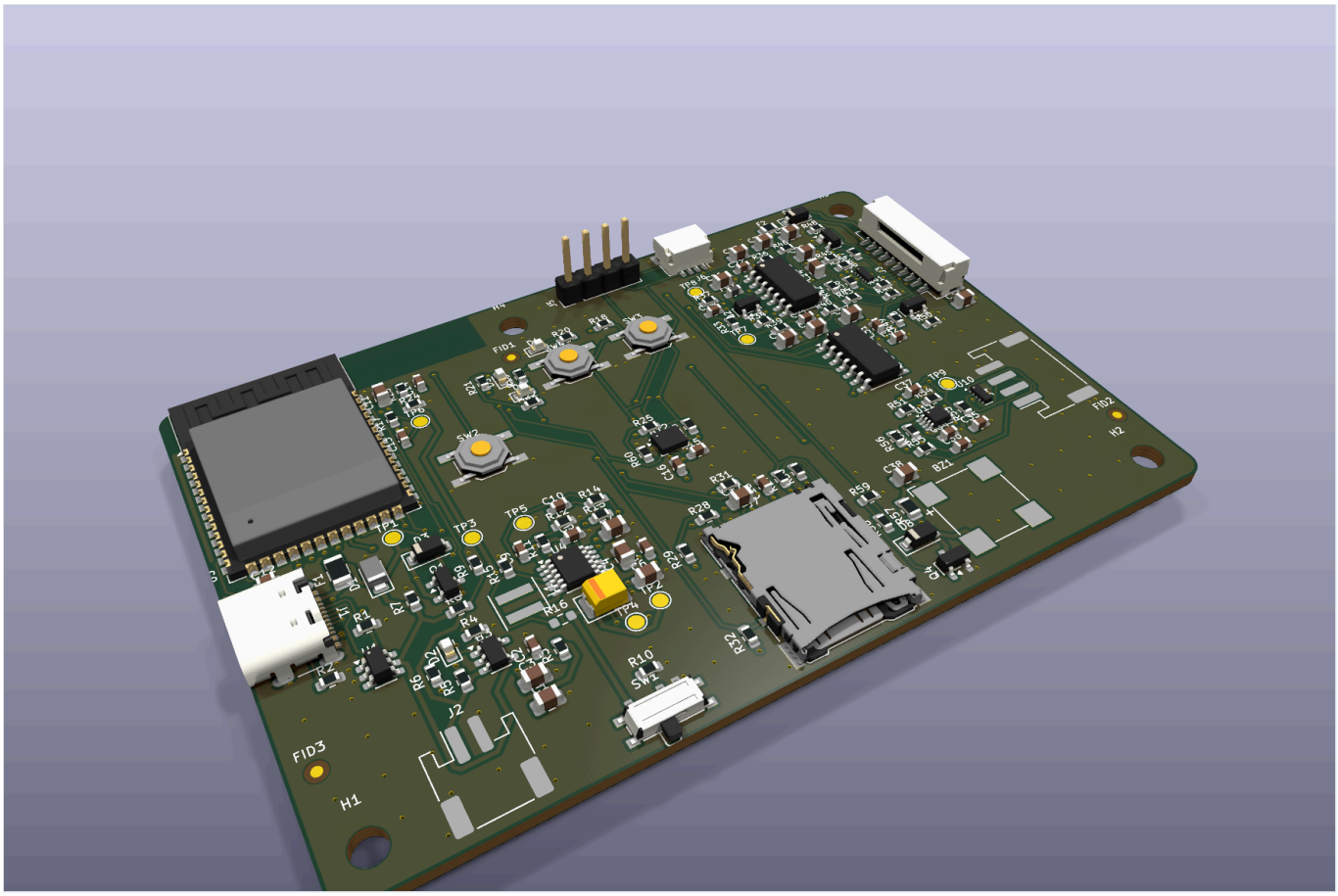
VW-MTC1 Controller Board: Design Report

Redesign of the thesis wire-maze controller: calculations, LTspice simulation, KiCad schematic and 4-layer PCB

Voltwyn | Document 20261019-VW-MTC1-Design-Report-Rev00 | 19 October 2026 | Author: Sajeevan (Saj) Veeriah



VW-MTC1 Rev A (KiCad 3D render)

1. Summary

The thesis prototype used an ESP32-WROOM-32 with four SS49E Hall sensors wired straight to the ESP32 ADC, an OLED, three LEDs and USB power from a host PC, with only an exponential moving average for noise control and no input protection. It reached 0 % drift at a 10 cm magnet pitch, but failed at 2.5 cm (-83 to -92 % drift), had 12.5-81 % direction misclassification depending on pitch, and 75.8 % availability in the reliability test (MTBF 6.3 s).

VW-MTC1 is a complete replacement controller. It adds proper signal conditioning and a precision ADC, battery operation with charging, protection on every external connection, and the features listed as future work in the thesis: on-device scoring and feedback, a custom integrated PCB, an IMU for 3D motion, haptic feedback, logging and remote connectivity. Every power and analogue block was sized by calculation and checked in LTspice; three value errors found by simulation were fixed before layout.

Requirement-to-result register

Request	Result	Status
Review the thesis	Section 2: original design, limits and how each is addressed	Done
New single controller board, more efficient	Buck-boost 91-94 % (SIM) replaces USB-only linear regulation; switched sensor rail; Burst Mode option	Done
More features and components	Precision ADC, filters, IMU, charger + battery, haptics, buzzer, maze contacts, microSD, Qwiic, ESD/TVS, buttons	Done
KiCad design	KiCad 10 project: 4-sheet schematic, 4-layer PCB; ERC 0 errors, DRC 0 errors, 0 unrouted, 0 schematic/PCB mismatches	Done
LTspice simulation	7 simulation sets (48 runs), 19 plots, results.json	Done
Completed board	Gerbers + drill, BOM, pick-and-place, STEP model ready for JLCPCB-class fabrication	Files done; not yet built or tested
Documentation, datasheet, report with images	Datasheet PDF, this report, schematic PDF, renders, layer plots	Done

Sellable with Voltwyn	Section 9: commercial and compliance pathway, open items	**Pathway defined; certification not started**

2. Review of the thesis controller

Thesis finding / limitation	Root cause (assessment)	VW-MTC1 response
Sensors wired directly into the ESP32 ADC; noise handled only by EMA (alpha 0.3) in firmware	No anti-alias filter; ESP32 ADC nonlinearity and supply noise	2nd-order Sallen-Key per channel (238 Hz) + MCP3204 12-bit ADC with VREF from the sensor rail (ratiometric). 9.3 dB SNR gain in SIM04
2 cm / 2.5 cm pitch: large drift and 62-81 % misclassification	Overlapping fields, threshold FSM, 500 Hz sampling, sensor geometry	2 kS/s per channel, cleaner signals for interpolation; recommended quarter-pitch sensor spacing (Section 8)
Reliability 75.8 % availability, manual resets	Noisy edges, no artefact rejection	Filtered + Schmitt-buffered inputs, watchdog-capable MCU, start/finish pads give automatic trial timing
USB-powered from a host PC only	No battery or power path	Li-ion charger, automatic USB/battery switchover, 3.3 V buck-boost
MATLAB on a PC needed for scoring	No local storage or processing	ESP32-S3 dual-core, microSD logging, OLED, BLE/Wi-Fi streaming
No protection on external connections	Hobby wiring	USBLC6, SMF5V0A, TPD4E05U06, PRTR5V0U2X ESD; PTC fuses on VBUS and motor
Future work: PCB, IMU, haptics, HMI, remote monitoring	-	All included on this board

Feature comparison

Item	Thesis board	VW-MTC1
MCU	ESP32-WROOM-32	ESP32-S3-WROOM-1-N8R2 (native USB, BLE 5, vector instructions)
Hall acquisition	ESP32 ADC, 500 Hz, no filter	MCP3204 12-bit ratiometric, 2 kS/s/ch, 232 Hz 2nd-order filter, backup ESP32 ADC path
Power	USB from PC, 3.3 V regulator	USB-C + 1S Li-ion, charger, power path, 1 A buck-boost
Feedback	OLED + 3 LEDs	OLED header, 3 LEDs, buzzer, handle ERM haptics
Error detection	Distance/direction only	Plus wire-contact, start and finish pads
Motion sensing	Hall only	Hall + 6-axis IMU
Data	USB to MATLAB	microSD, USB CDC, BLE / Wi-Fi
Protection	None	ESD on all connectors, VBUS TVS, PTC fuses
Expansion	-	Qwiic I2C (e.g. VL53L1X LiDAR, as planned in the thesis)

3. Architecture

USB-C input
CC 5.1k Rd, 0.75 A PTC, SMF5V0A TVS, USBLC6 ESD

Li-ion charger
LTC4054, 400 mA, thermal regulation

Power path
B5819W OR-ing + AO3401A load share

3.3 V buck-boost
LTC3536, 1 A, 1.2 MHz, 91-94 % eff.

ESP32-S3-WROOM-1
Wi-Fi + BLE 5, native USB, 8 MB flash

Handle interface
JST-GH 10: 4x SS49E, I2C, ERM motor

Hall front end
4x Sallen-Key 232 Hz, MCP6004

12-bit ADC
MCP3204, ratiometric VREF

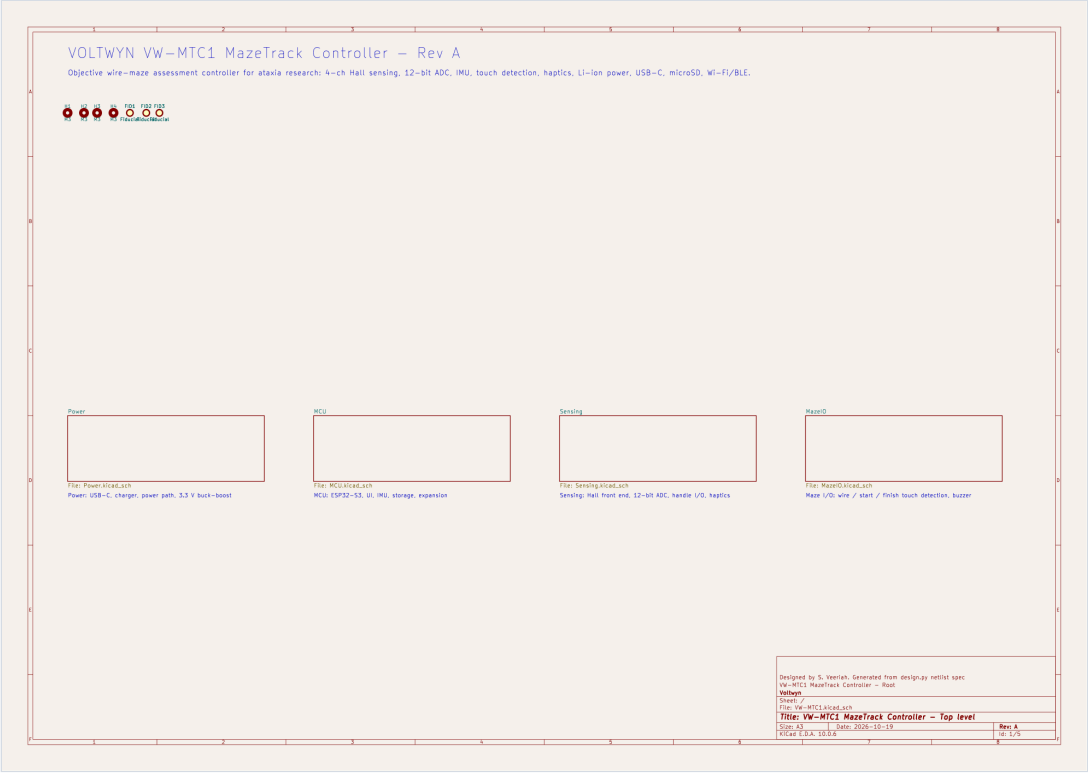
Maze inputs
Wire / start / finish, 9 uA max, Schmitt

IMU
ISM330DHCX 6-axis

Storage
microSD (SDMMC)

UI
OLED header, 3 LEDs, 3 buttons, buzzer

Expansion
Qwiic I2C (e.g. LiDAR)



Top-level schematic (4 hierarchical sheets)

4. Design calculations

4.1 USB-C input and protection

CC1/CC2 each have a 5.1 kohm Rd to GND, so any USB-C source supplies default USB power. A 0.75 A hold PTC (1206L075) limits fault current, an SMF5V0A TVS (5 V stand-off) clamps VBUS surges and a USBLC6-2SC6 protects D+/D-. The D+/D- crossover between the A and B pin rows is hand-routed with 0.5 mm vias before autorouting.

4.2 Li-ion charger (LTC4054ES5-4.2)

Quantity	Working	Result
Charge current	I = 1000 x V_PROG / R_PROG = 1000 x 1 V / 2.49 k	402 mA (SIM: 407.9 mA)
Die power, worst case CC	P = (5.0 - 3.0) x 0.40	0.80 W
Junction temp, 4-layer (theta-JA 80 C/W, datasheet table 2)	Tj = 25 + 0.80 x 80	89 C: below the 120 C regulation point
Typical (VBAT 3.7 V)	Tj = 25 + 0.52 x 80	67 C
Hot-plug damping	1.5 ohm in series with 10 uF (datasheet recommendation)	Peak 5.00 V vs 7.96 V without (SIM02b)
Charge time, 2000 mAh cell (estimate)	~1.1 x C / I	~5.5 h

4.3 Power path

USB feeds VSYS through a B5819W Schottky (1 A). The battery feeds VSYS through an AO3401A P-MOSFET whose gate is VBUS: with USB present VGS is about +0.4 V (off); when USB is removed R7 pulls the gate to 0 V and the FET turns on (VGS = -VBAT). SIM03 showed the original 100k pull-

down never discharged VBUS fast enough, so the FET stayed off and VSYS ran on the body diode at 2.95 V. R7 is now 2.2 kohm (2.3 mA, 11 mW while on USB) and Q1 turns on 5.5 ms after unplug. C39 (100 uF polymer tantalum) cut the replug overshoot on +3V3 from +6.4 % to +3.3 %.

4.4 3.3 V buck-boost (LTC3536)

Quantity	Working	Result
Output voltage	VOUT = 0.6 x (1 + 1M / 221k)	3.315 V
Frequency	RT tied to VIN (datasheet)	1.2 MHz
Inductor ripple (buck, VIN 4.6 V)	dI = VOUT (1 - VOUT/VIN) / (L f) = 3.3 x 0.283 / (4.7u x 1.2M)	0.17 A p-p
Compensation	ADI reference: 47 pF + 6.49k feed-forward, 220 pF + 49.9k FB-VC	Stable with ceramic output, no ringing (SIM01b)
Mode	R15 0 ohm: MODE low = PWM (lowest noise); fit R16 instead for Burst Mode	Burst at 20 mA: 79 % (SIM01d)
Why buck-boost	Li-ion 3.0-4.2 V spans 3.3 V; an LDO would drop out below ~3.5 V	Uses the full cell capacity

4.5 Power budget (estimate, not measured)

Load	Typical current at 3.3 V	Source
ESP32-S3 active, BLE streaming	60-100 mA (Wi-Fi TX peaks ~350 mA)	Typical module figures, to verify on hardware
4x SS49E Hall sensors	~24 mA	~6 mA each (Honeywell typical)
OLED 0.96 inch	10-20 mA	Content dependent
IMU, op-amp, ADC	~2 mA	Datasheet typicals
LEDs, microSD (average)	5-15 mA	Duty dependent
Total	~100-160 mA (~0.33-0.53 W)	

From a 3.7 V cell at 92 % efficiency, the battery current is about 100-155 mA, so a 2000 mAh cell gives roughly 13-20 hours of active use (estimate). Switching off +3V3_H between sessions removes the ~24 mA Hall load.

4.6 Hall sensor front end

Quantity	Working	Result
SS49E sensitivity at 3.3 V	1.4 mV/G at 5 V, ratiometric: 1.4 x 3.3 / 5	~0.92 mV/G
ADC LSB	3.3 V / 4096	0.806 mV, ~0.87 G
Filter cut-off	fc = 1 / (2 pi R sqrt(C1 C2)) = 1 / (2 pi x 10k x sqrt(100n x 47n))	232 Hz (SIM 238 Hz)
Q	Q = 0.5 sqrt(C1 / C2)	0.73 (Butterworth-like)
Signal bandwidth needed	Magnet pass ~20 ms at 50 cm/s; tremor < 15 Hz	< 50 Hz: passband flat (+0.01 dB)
Sampling budget	24 SPI clocks x 4 ch at 1 MHz = 96 us per frame	19 % of the 500 us frame at 2 kS/s
Ratiometric benefit	Sensor supply and ADC VREF share +3V3_H	Supply drift and ripple cancel to first order
Handle absent	1 Mohm pull-down per input	Reads ~0 V instead of the 1.65 V null

4.7 Maze contact inputs

The ring on the handle is at 0 V. Each maze input has a 10 kohm series resistor, a 360 kohm pull-up and 27 nF to ground, followed by a 74LVC3G17 Schmitt buffer. The worst-case DC current through a person bridging wire and ring is 3.3 V / 370 kohm = 8.9 uA, within the 10 uA IEC 60601-1 type BF patient auxiliary current target (design target, not a certification). The values were changed after SIM05 showed the first choice (100k / 1M / 10 nF) chattered under mains coupling.

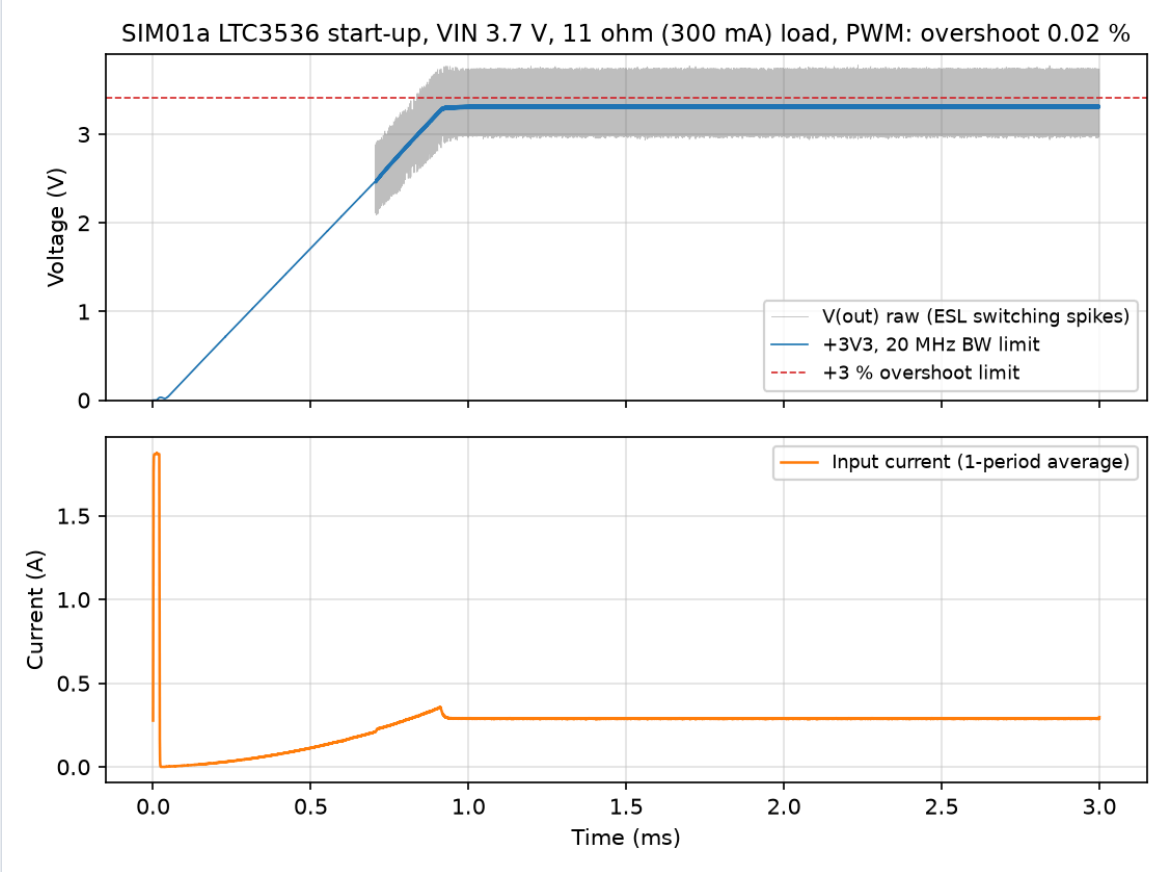
4.8 Other blocks

Block	Calculation	Result
Status LEDs	(3.3 - ~2.0 V) / 470 ohm	~2.8 mA each
I2C0 pull-ups 4.7k	t_r = 0.85 x 4.7k x 75 pF	0.30 us: meets 400 kHz up to ~75 pF
Handle I2C pull-ups 2.2k	t_r = 0.85 x 2.2k x 150 pF (1 m cable)	0.28 us
Battery monitor	1M / 1M divider + 100 nF	2 uA drain; VBAT = 2 x Vadc
Haptic ERM	20 kHz PWM from VSYS	Duty 76 % / 55 % for 3.0 V average (SIM06)
Buzzer	22 ohm + 10 uF reservoir from +3V3	9.1 mV rail ripple (SIM07)

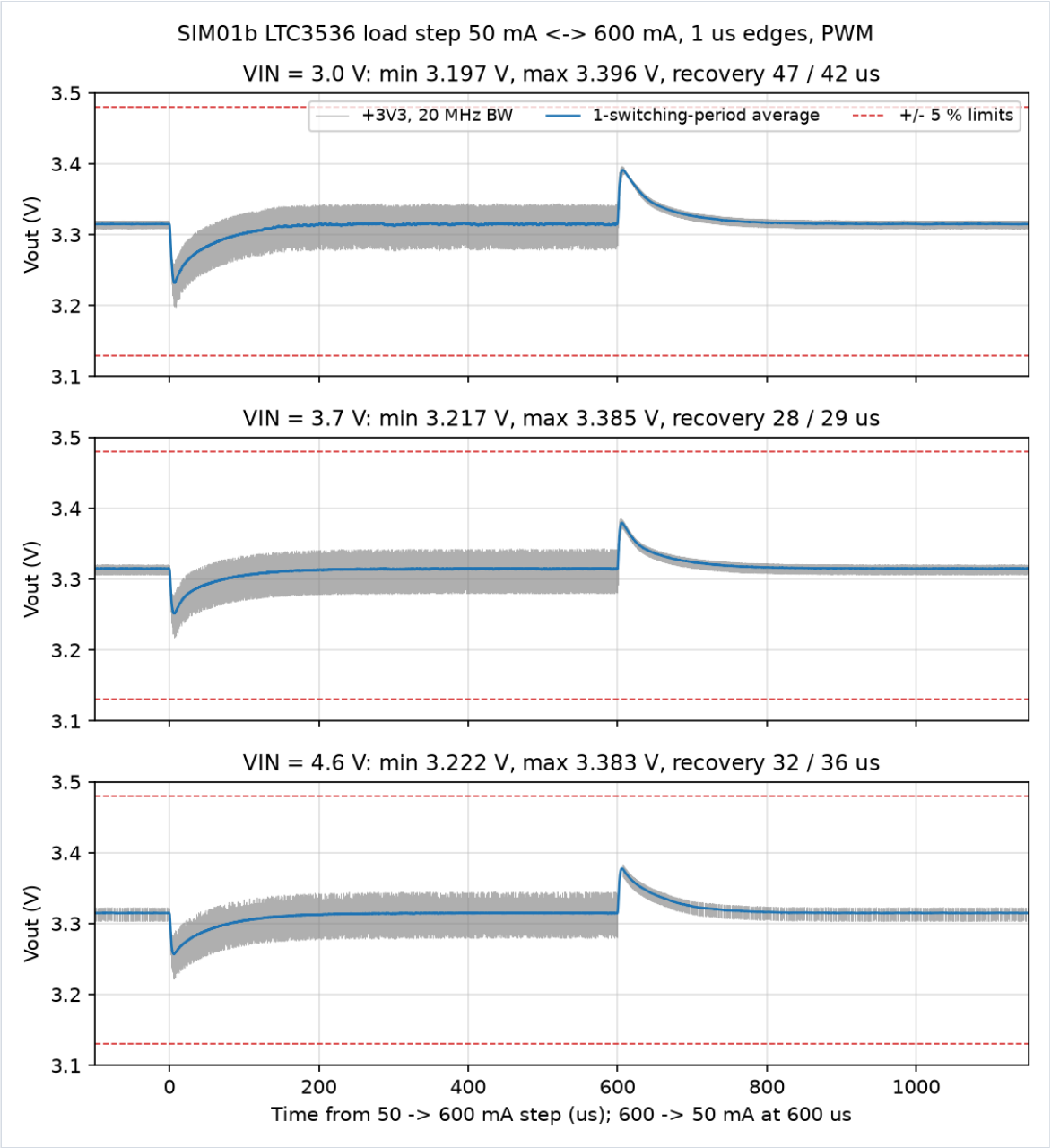
5. LTspice simulation results

All simulations ran in LTspice 26.1.1 batch mode using ADI vendor models for the LTC3536 and LTC4054. Netlists, logs and the schematic (.asc) files are in `Simulation/LTspice`. Simulation reference designators differ from the board. Mapping: L1 = L1; C30/C31 = C6/C7; C28/C29 = C4/C5; R30/R31 = R11/R12; C32/R32 = C9/R13; C33/R33 = C10/R14; R21 = R4; C20 + R20 = C1 + R3; D40 = D3; R43 = R7; C42 = C39; Hall filter R50-R52, C51, C52 = R35-R37, C23, C24 (channel 1); touch RS/RPU/CF = R51/R52/C34 (wire channel); ERM Q2 = Q3.

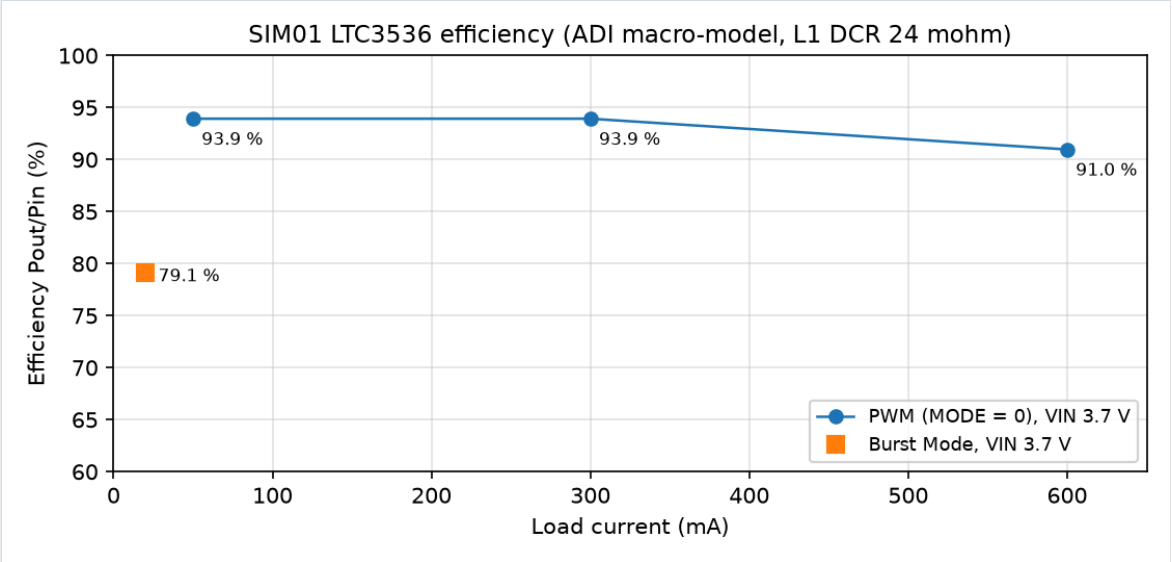
Sim	Check	Result	Limit	Status
SIM01	+3V3 start-up overshoot	0.38 %	< 3 %	PASS
SIM01	Load step 50-600 mA, VIN 3.0 / 3.7 / 4.6 V	3.197-3.396 V	3.13-3.48 V	PASS
SIM01	Ripple, 300 mA	37 mV p-p	-	info
SIM01	Efficiency 50 / 300 / 600 mA	93.9 / 93.9 / 91.0 %	-	info
SIM02	CC charge current	407.9 mA	400 mA +/-7 %	PASS
SIM02	Float voltage	4.217 V	4.2 V +/-1 %	PASS
SIM02	Termination current	28.8 mA	~40 mA	Model behaviour (terminates at V_PROG 80 mV); no board change
SIM03	+3V3 during unplug / replug (final values)	3.262-3.419 V	+/-5 %	PASS
SIM04	Filter -3 dB, Q, group delay at 10 Hz	238 Hz, 0.729, 0.95 ms	< 2 ms delay	PASS
SIM04	ADC sample settling (20 pF)	0.64 us	< 1.5 us	PASS
SIM05	False triggers, 50 / 100 pF mains coupling (final values)	None	None	PASS
SIM05	Two touches 5 ms apart	Merged into one pulse	Separate	Open: firmware counts edges
SIM06	MOSFET drain peak at turn-off	3.98 / 4.90 V	< 20 V	PASS
SIM07	Buzzer drain peak / rail ripple	4.23 V / 9.1 mV	-	PASS



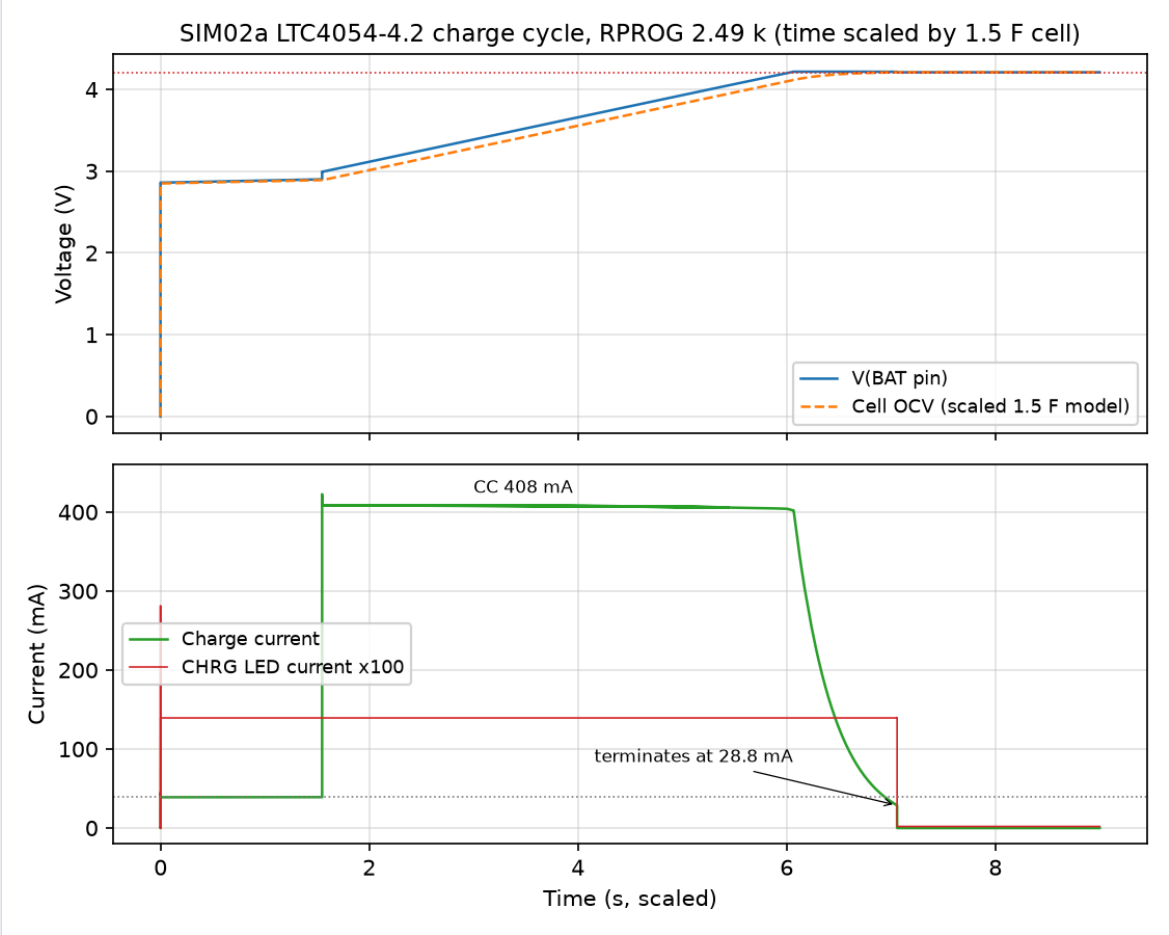
SIM01a: buck-boost start-up



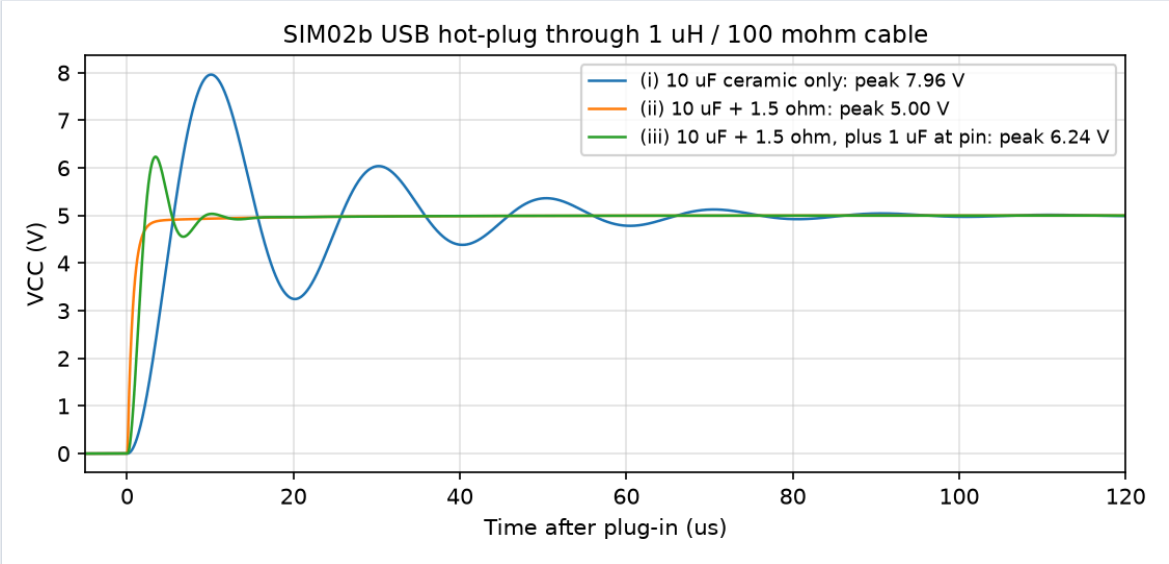
SIM01b: 50-600 mA load steps at three input voltages



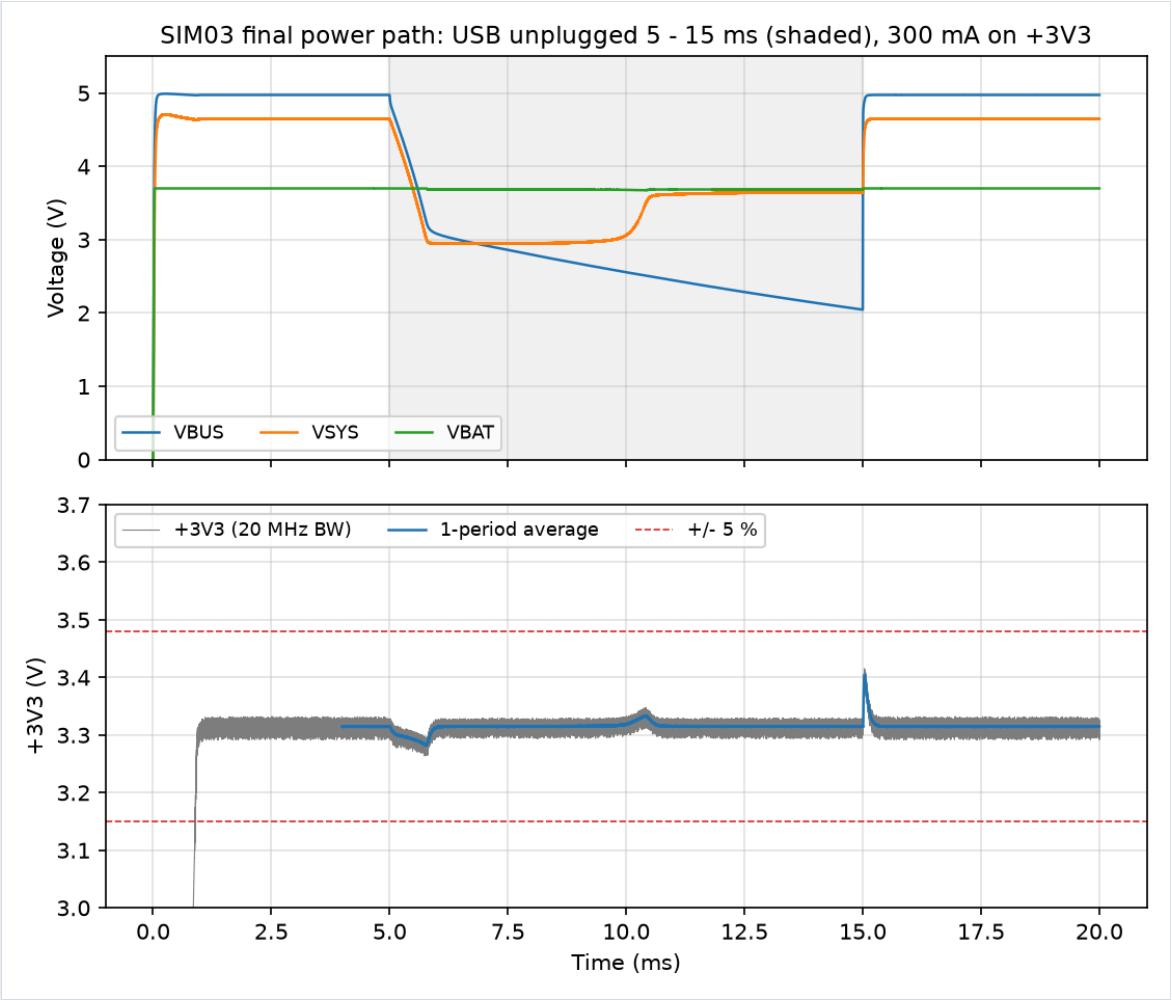
SIM01c: efficiency



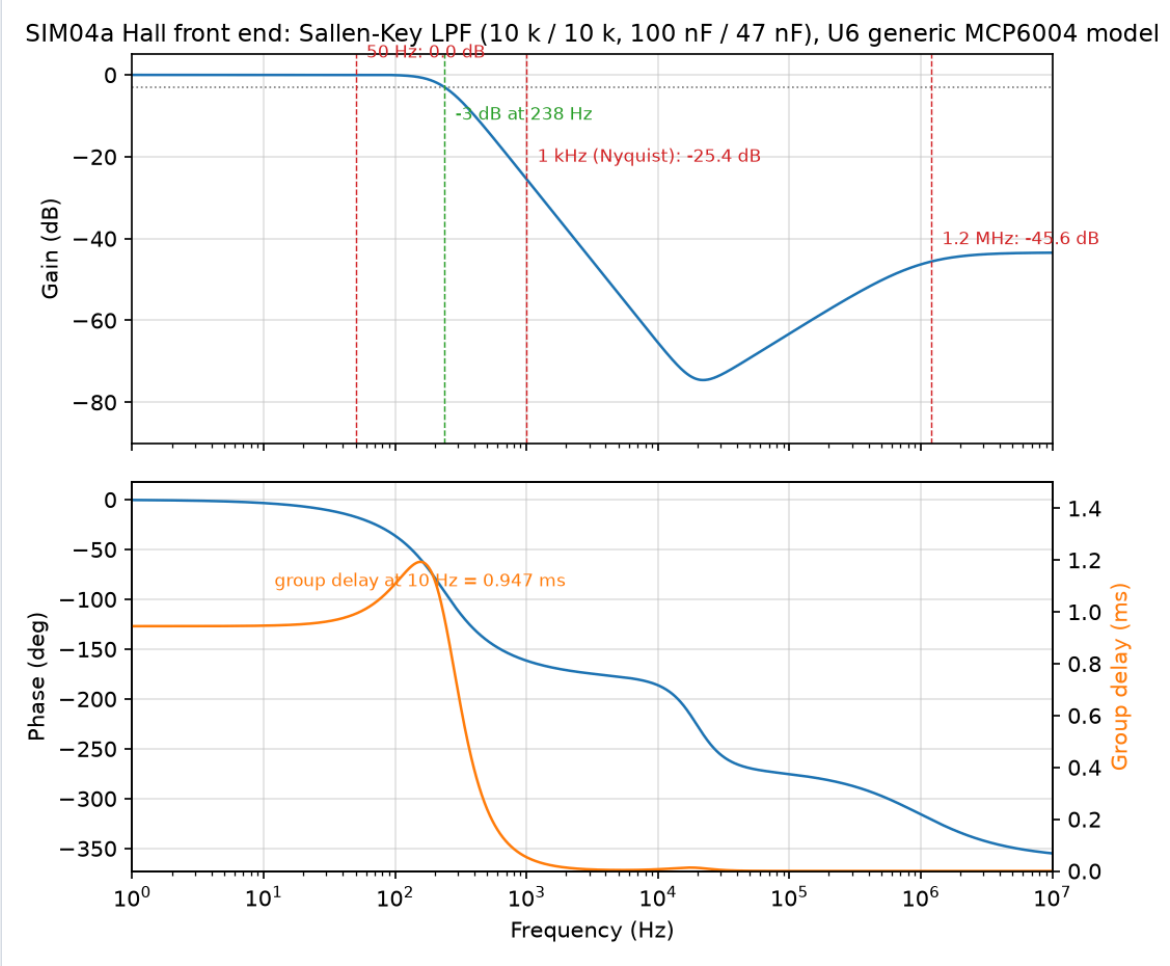
SIM02a: charger trickle, CC and CV phases



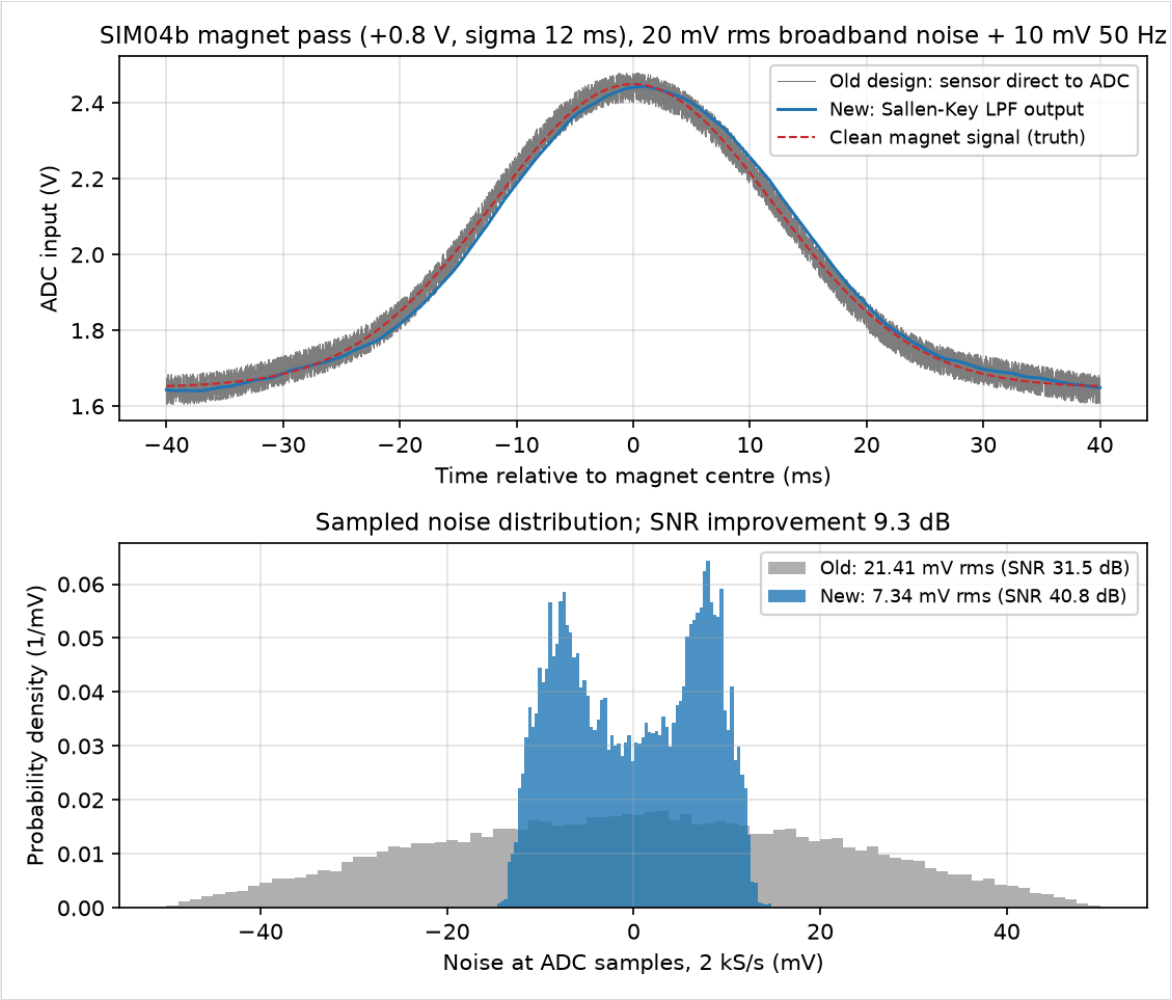
SIM02b: USB hot-plug with and without damping



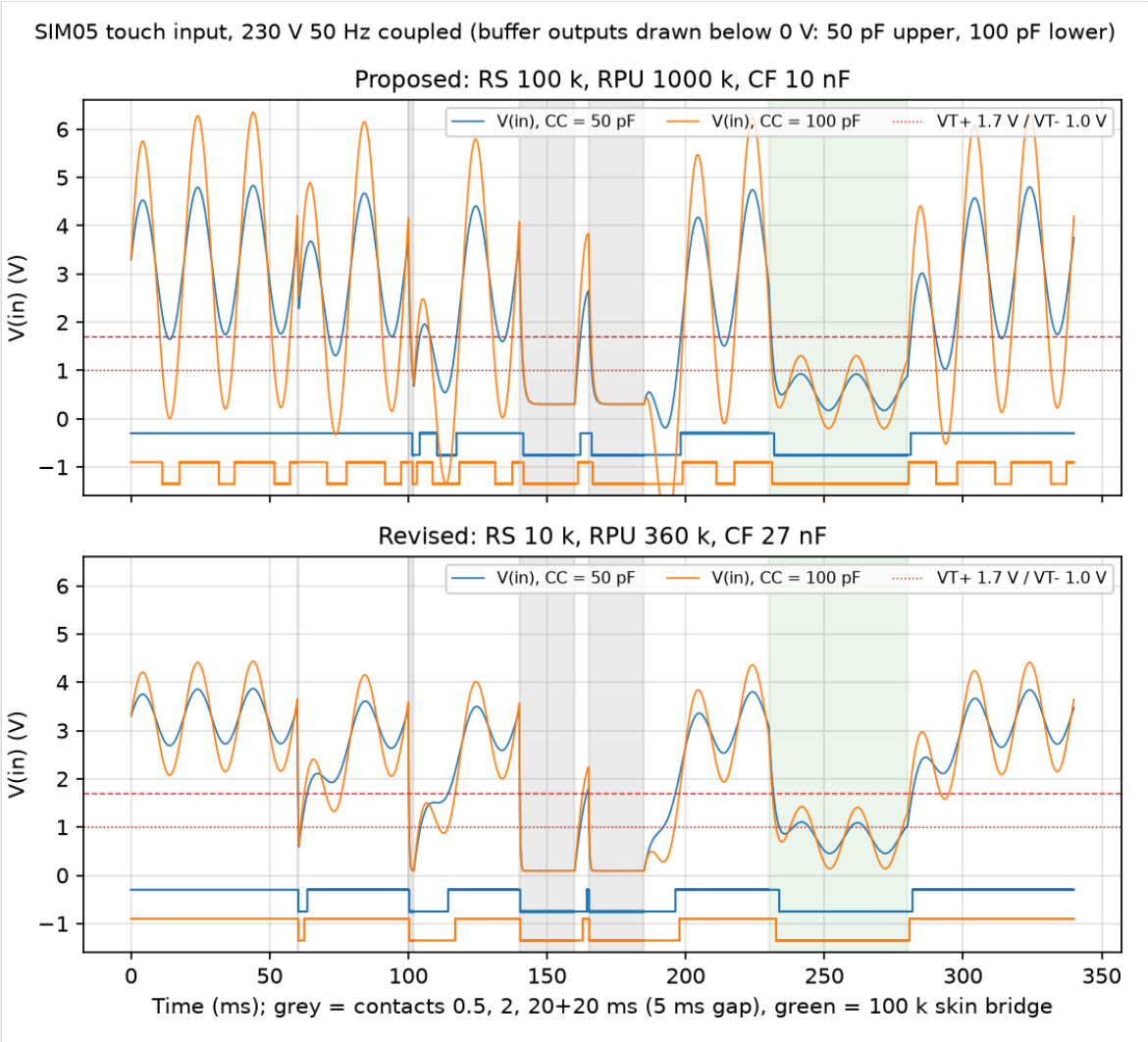
SIM03: USB unplug/replug with final values



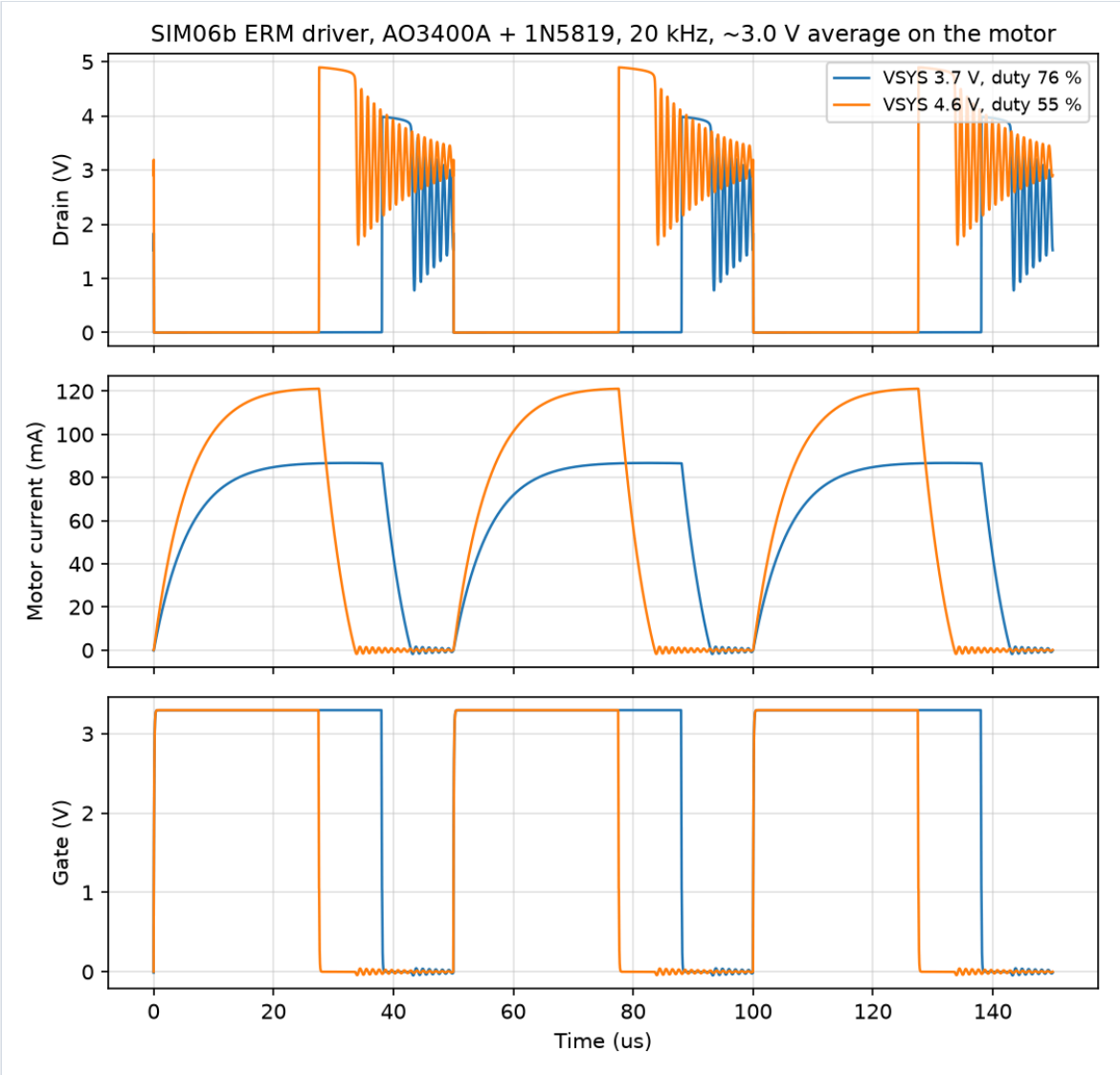
SIM04a: Hall filter Bode plot



SIM04b: thesis design vs new front end



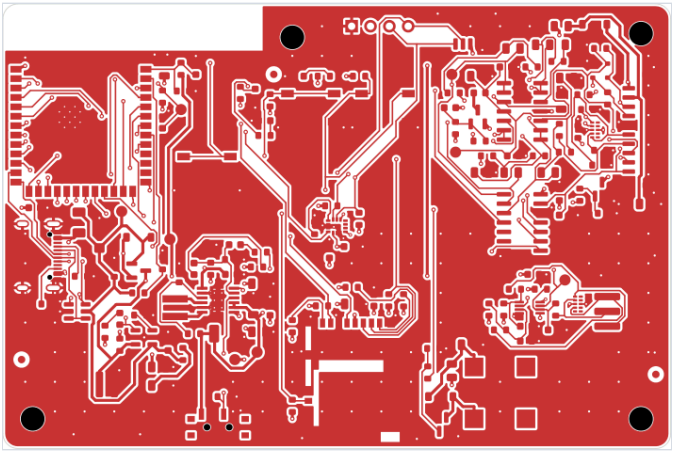
SIM05: maze contact input under mains coupling



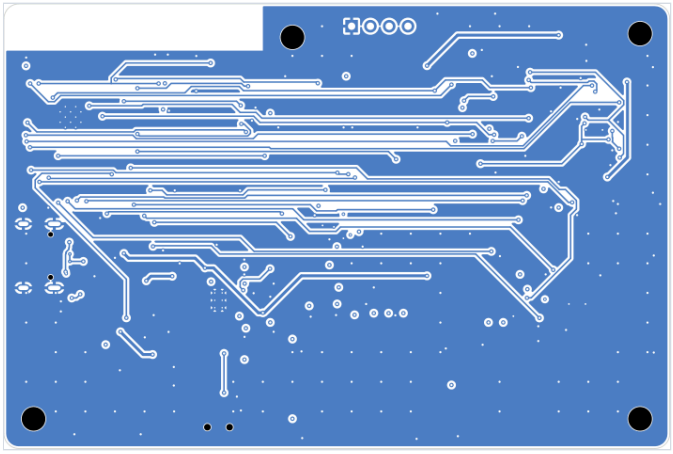
SIM06: haptic ERM driver

6. PCB design

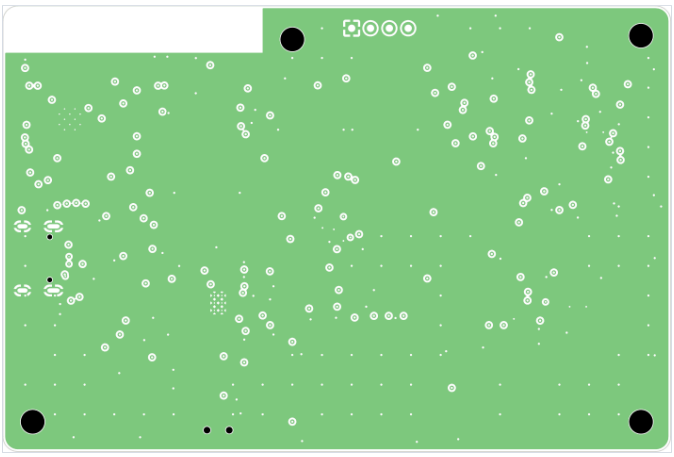
Item	Value
Tool	KiCad 10.0.6 (schematic, PCB, DRC); Freerouting 2.0.1 autorouter with hand-routed critical nets
Stackup	L1 signal + GND pour / L2 GND plane / L3 +3V3 plane / L4 signal + GND pour, 1.6 mm
Rules	0.15 mm min clearance (0.1 mm inside fine-pitch packages), 0.2 mm signal, 0.4 mm power, 0.6/0.3 mm vias
Routing statistics	1238 track segments, 3.3 m copper (top 1.72 m, bottom 1.58 m), 317 vias including GND stitching
Power stage	LTC3536 with inductor beside the SW pins, exposed pad with thermal vias, SGND/PGND tied at the pad
Antenna	ESP32-S3 at the board edge, 35 x 6.3 mm keepout on all layers plus the module footprint rule area
Plane connections	139 GND / +3V3 pads fanned out to the inner planes by script before routing
Verification	ERC: 0 errors (1 warning: unused IMU SDx tied to GND per ST). DRC: 0 errors, 0 unconnected, 0 schematic-parity issues; 219 warnings (silkscreen overlap, small isolated copper islands)



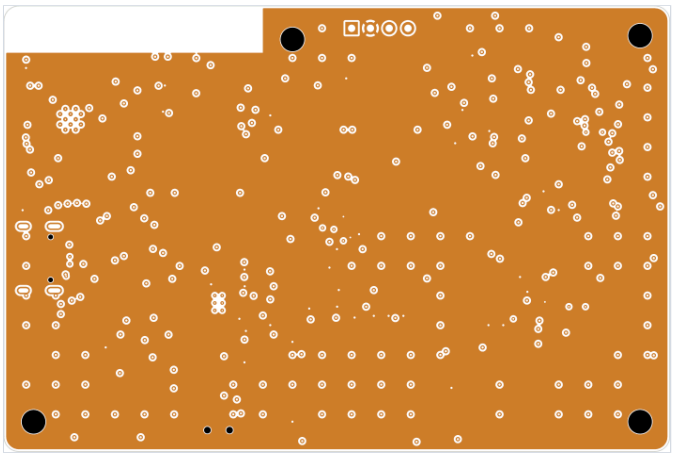
Top copper (F.Cu)



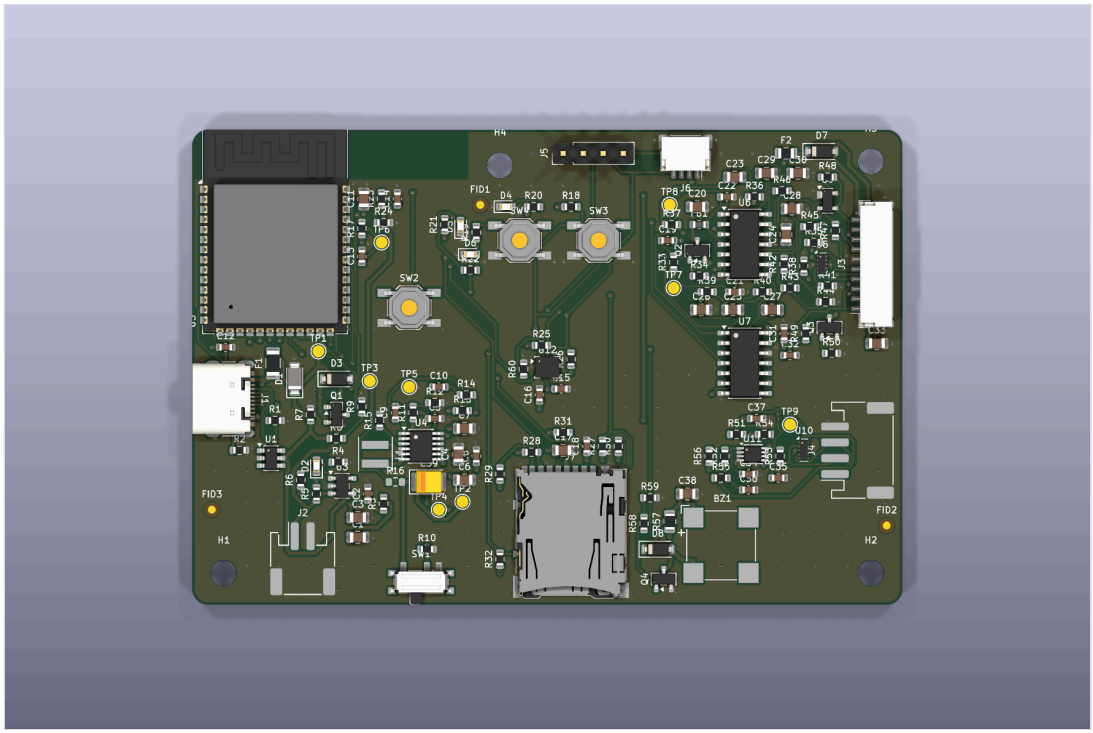
Bottom copper (B.Cu, viewed from top)



Inner 1: GND plane

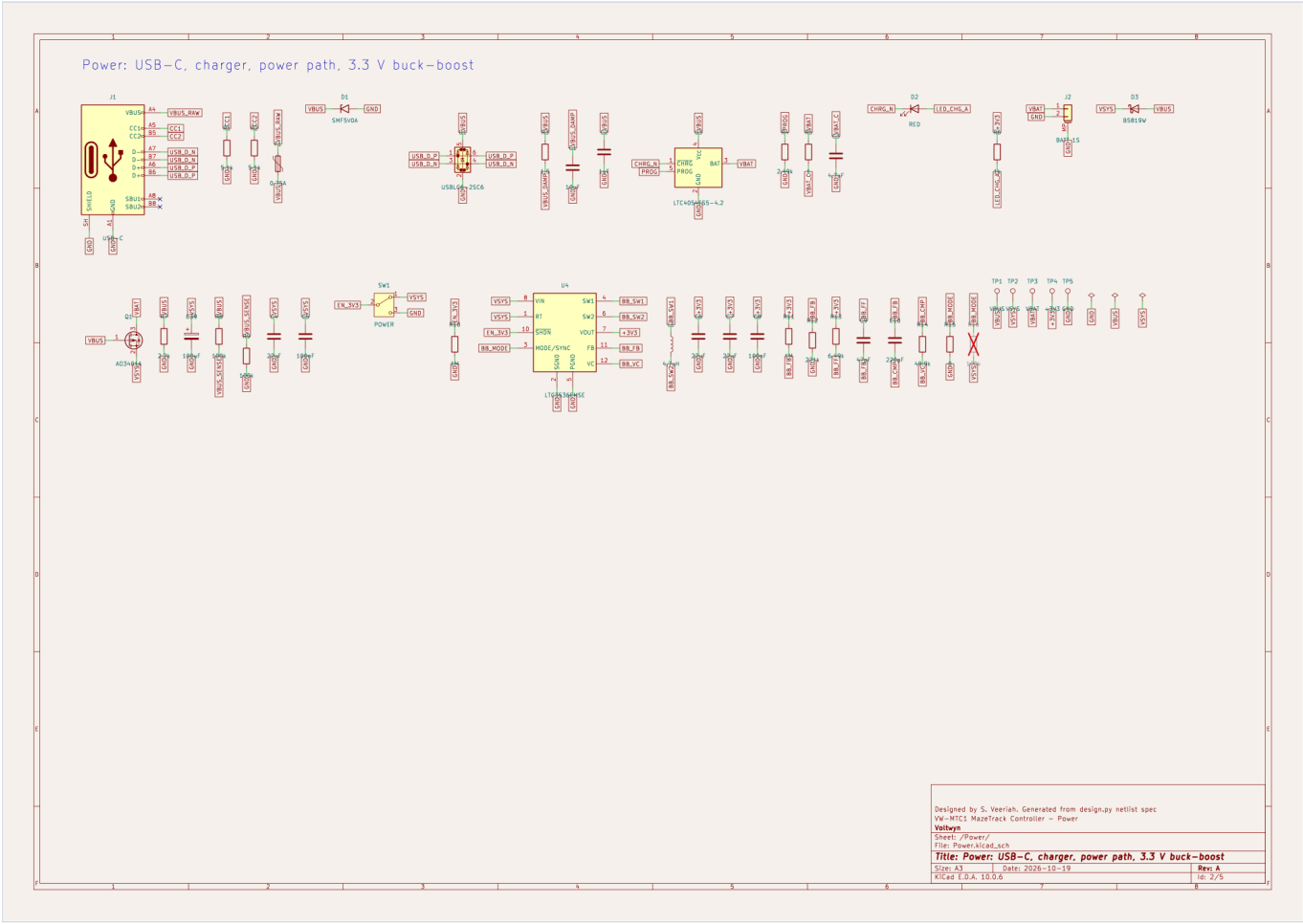


Inner 2: +3V3 plane

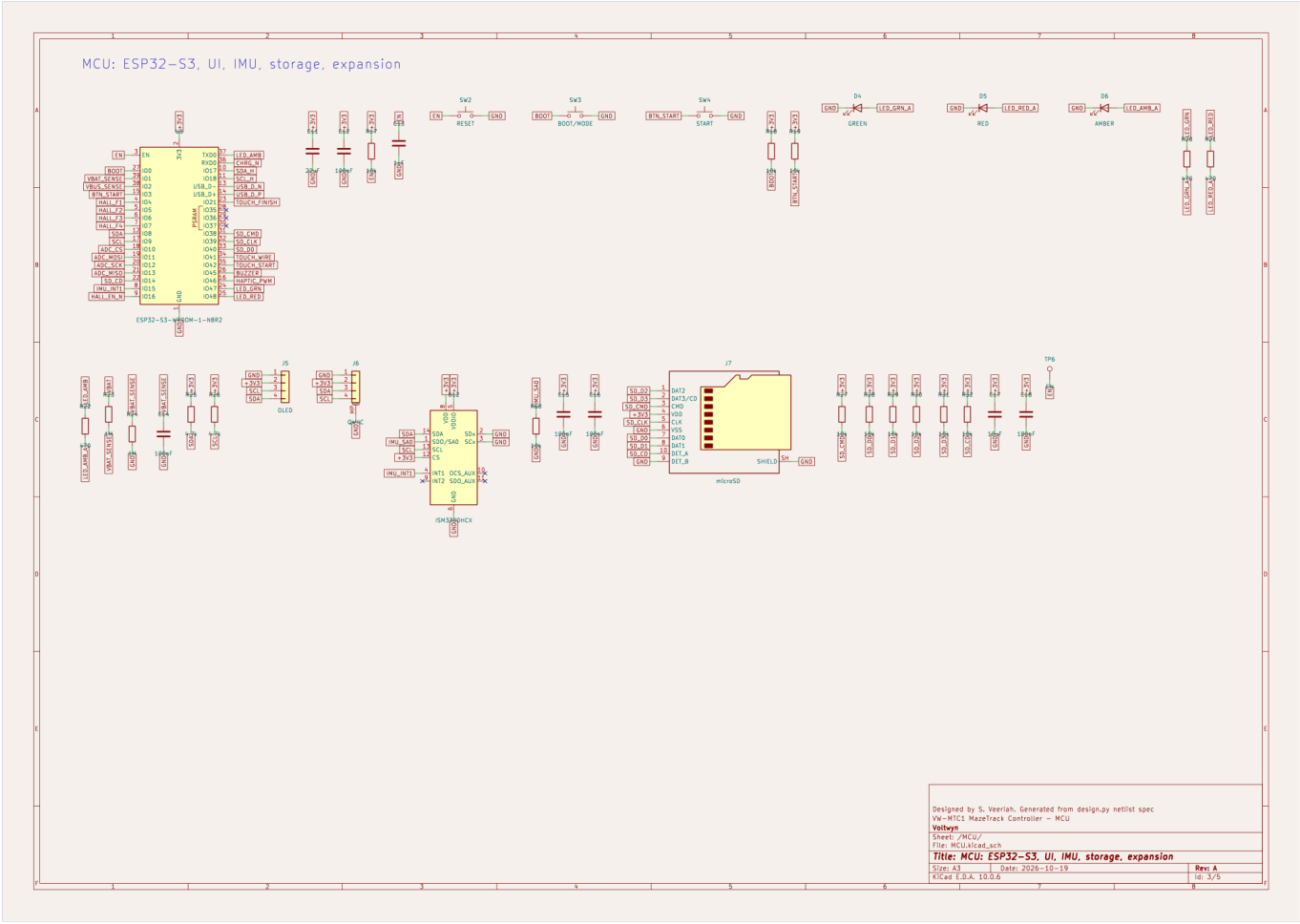


Top render

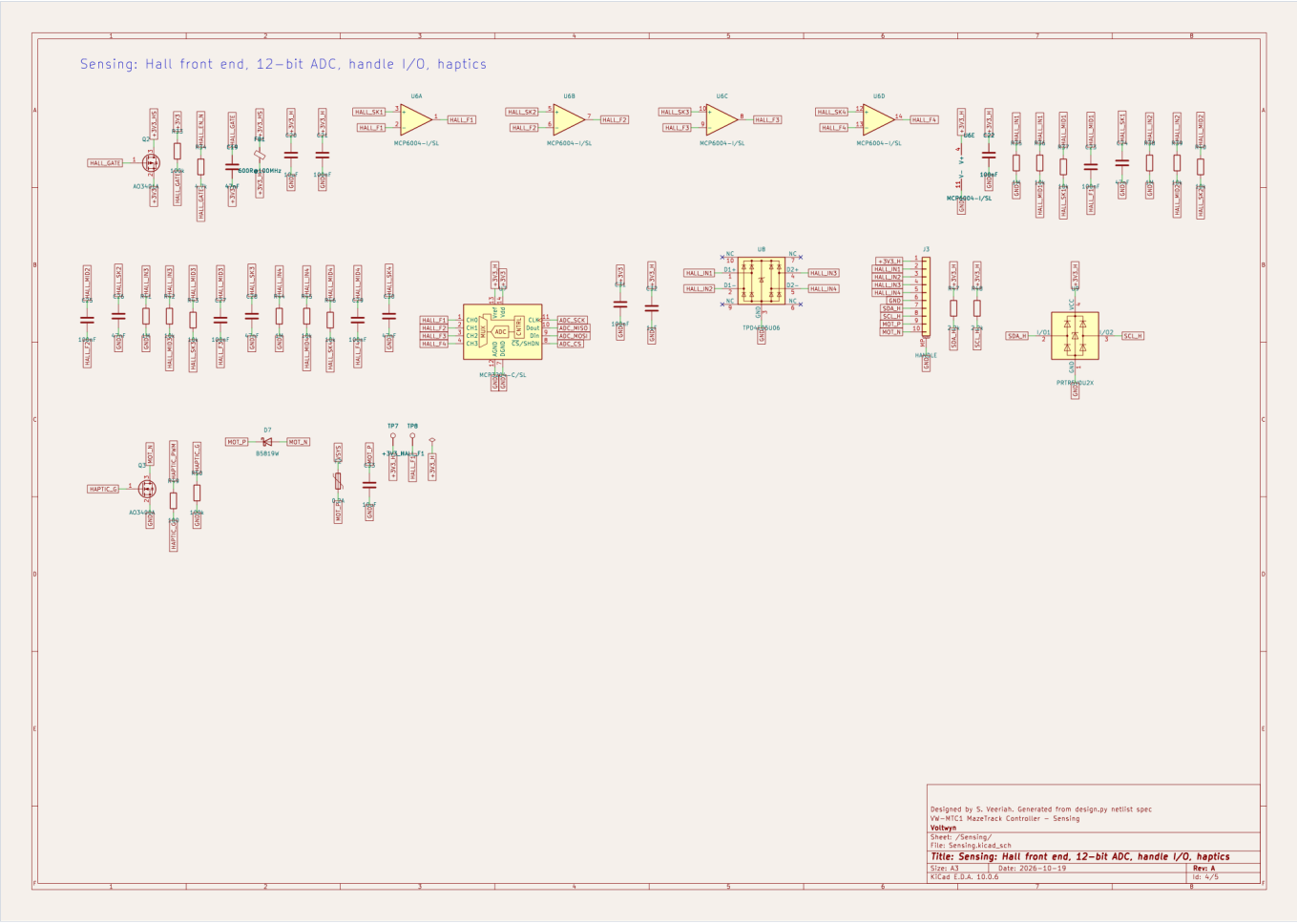
Schematic sheets



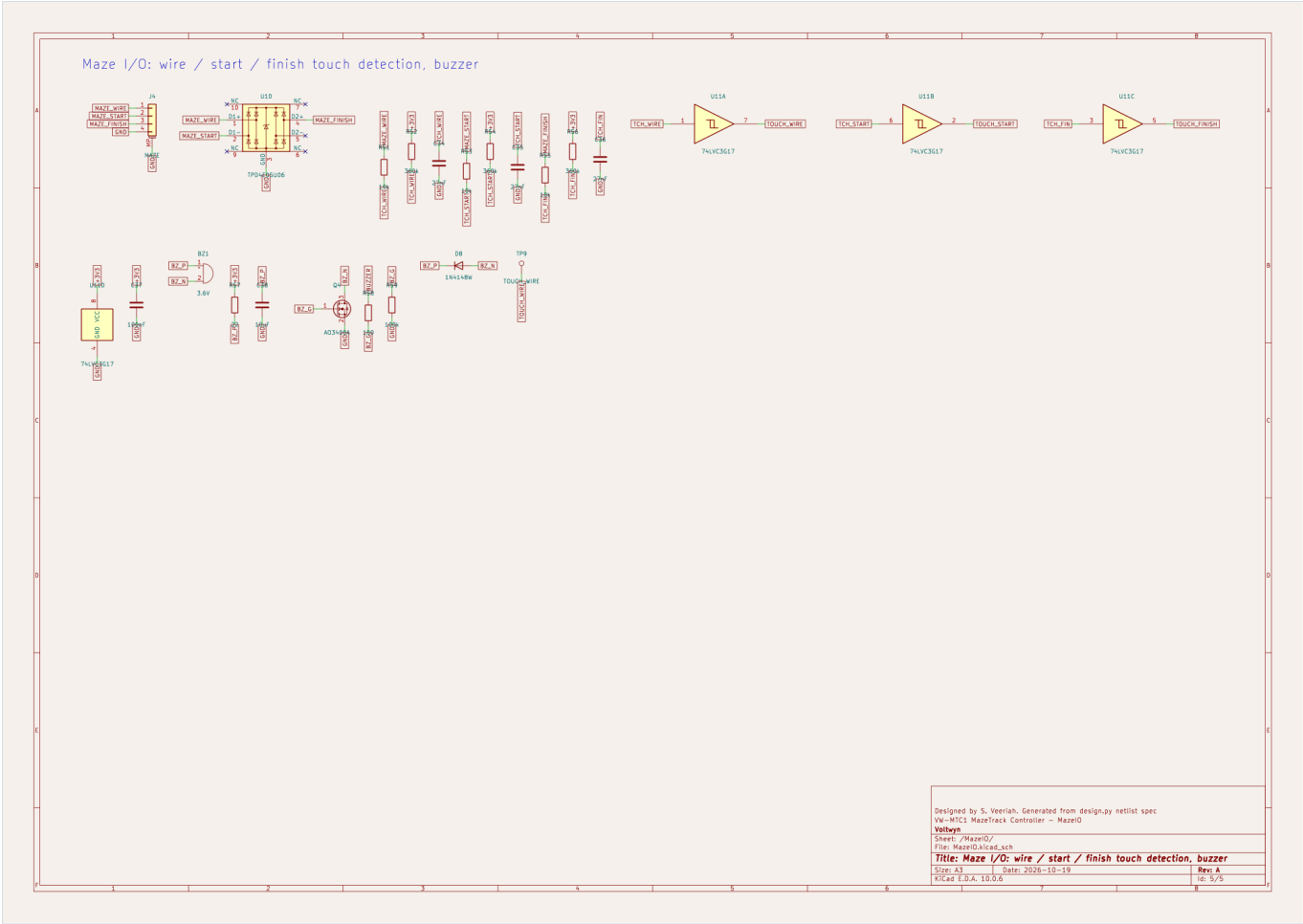
Power: USB-C, protection, charger, power path, buck-boost



MCU: ESP32-S3, UI, IMU, microSD, expansion



Sensing: Hall front end, ADC, handle interface, haptics



Maze I/O: contact inputs and buzzer

7. Manufacturing package

File	Purpose
Hardware/KiCad/VW-MTC1.kicad_pro (+ .kicad_sch x5, .kicad_pcb, .kicad_dru, Voltwyn.kicad_sym)	Editable KiCad 10 project
Hardware/Fabrication/20261019-VW-MTC1-Gerbers-Drill-Rev00.zip	Gerber X2 (11 layers) + Excellon drill + drill maps + job file
Hardware/Fabrication/20261019-VW-MTC1-BOM-Rev00.csv	Bill of materials: 64 lines, 138 placed parts
Hardware/Fabrication/20261019-VW-MTC1-PickPlace-Rev00.csv	Top-side pick-and-place (DNP excluded)
Hardware/Fabrication/20261019-VW-MTC1-3D-Model-Rev00.step	Mechanical model for enclosure design
Hardware/Reports	ERC and DRC reports

Order as a 4-layer, 1.6 mm, 1 oz board, minimum 0.1 mm space / 0.15 mm track, 0.2 mm drill, ENIG finish recommended for the LGA IMU and fine-pitch parts. The LTC3536 MSOP-12 exposed pad and the ISM330DHCX LGA-14 need a stencil (0.12 mm) and reflow. Prices and stock were not quoted in this revision.

8. Firmware and system recommendations

- Sample the MCP3204 at 2 kS/s per channel on a hardware timer; subtract the fixed 0.95 ms filter delay from event times.
- Place the handle sensors a quarter magnet pitch apart (for example 2.5 cm for a 10 cm pitch) so paired sensors give quadrature signals. Direction then comes from phase, and position from atan2 between magnets, instead of threshold counting. This targets the 2-2.5 cm pitch failures in the thesis.
- Count maze contacts by falling edges (not pulse width); debounce 2 ms in firmware.
- Use the start and finish pads to time trials automatically, and compute the thesis composite score on the device; show it on the OLED and log raw data to microSD.
- Power down +3V3_H (GPIO16 high) between trials; fit R16 instead of R15 if Burst Mode is wanted for long idle periods.
- Fuse the IMU and Hall data for 3D handle path and tremor metrics; stream over BLE for clinician dashboards.

9. Commercialisation pathway for Voltwyn

Area	What is needed	Status
Intended use	Sell first as a research / education / rehabilitation-game instrument with no diagnostic claims	Recommended
Medical device route (Australia)	If marketed for diagnosis or monitoring: TGA classification, ARTG inclusion, QMS (ISO 13485), risk management (ISO 14971), IEC 60601-1 and IEC 60601-1-2 testing	Not started
EMC and radio	RCM marking via ACMA requirements; the ESP32-S3 module carries its own radio approvals (confirm the AU/NZ certificates)	Not started
Battery	Use a certified protected cell (IEC 62133); UN38.3 documentation for shipping	Cell not yet selected
Product safety	Enclosure, labelling, user manual, cleaning instructions for patient-contact parts (handle ring)	Not started
Production	Assembled pilot run, bring-up tests, calibration jig, functional test firmware	Next step

10. Verification summary and open items

Check	Result
Schematic ERC (KiCad 10.0.6)	0 errors, 1 benign warning
PCB DRC with schematic parity	0 errors, 0 unconnected, 0 parity issues; 219 cosmetic warnings
LTspice	48 runs; all final-value checks pass except the items below
Datasheet pin checks	LTC3536 MSOP-12 pinout and LTC4054 pin functions checked against the ADI datasheets; other parts use KiCad library symbols (not individually checked against PDFs)

Open items before production

- Hardware not yet built or measured. All performance figures are calculated or simulated.
- USB inrush with the 100 uF on VSYS was not checked against USB attach limits (the simulation model used an ideal source).
- Two maze touches closer than ~15 ms merge into one pulse; firmware must count edges.
- Charger termination simulated at 28.8 mA vs ~40 mA expected (vendor model behaviour; verify on hardware).
- The op-amp and MOSFETs used fitted generic models; ceramic DC-bias derating was estimated.
- Silkscreen reference overlaps (DRC warnings) should be tidied before release; some 3D models (inductor, buzzer) are missing from the KiCad library.
- Battery JST-PH polarity varies between vendors; confirm the cell lead polarity (pin 1 = +).
- BOM pricing and stock not quoted; lifecycle audit not performed.

Appendix A. Bill of materials

Designators	Value	MPN	Manufacturer	Qty	
BZ1	3.6V	CMT-8504-100-SMT-TR	CUI Devices	1	
C1,C17,C20,C33,C38	10uF	GRM21BR61E106KA73L	Murata	5	
C2,C13,C32	1uF	GRM188R61E105KA12D	Murata	3	
C3	4.7uF	GRM21BR61C475KA88L	Murata	1	
C4,C6,C7,C11	22uF	GRM21BR61A226ME44L	Murata	4	
C5,C8,C12,C14,C15,C16,C18,C21,C22,C31,C37	100nF	GRM188R71H104KA93D	Murata	11	
C9	47pF	GRM1885C1H470JA01D	Murata	1	
C10	220pF	GRM1885C1H221JA01D	Murata	1	
C19	47nF	GRM188R71H473KA61D	Murata	1	
C23,C25,C27,C29	100nF	GCM21B5C1H104JA16L	Murata	4	
C24,C26,C28,C30	47nF	GRM2195C1H473JA01D	Murata	4	
C34,C35,C36	27nF	GRM188R71H273KA01D	Murata	3	
C39	100uF	T520B107M006ATE070	KEMET	1	
D1	SMF5V0A	SMF5V0A	Littelfuse	1	
D2,D5	RED	150060RS75000	Wurth	2	
D3,D7	B5819W	B5819W	Diodes Inc	2	
D4	GREEN	150060GS75000	Wurth	1	
D6	AMBER	150060AS75000	Wurth	1	
D8	1N4148W	1N4148W-7-F	Diodes Inc	1	
F1	0.75A	1206L075THYR	Littelfuse	1	
F2	0.2A	0805L020WR	Littelfuse	1	
FB1	600R@100MHz	BLM18PG601SN1D	Murata	1	
J1	USB-C	USB4105-GF-A	GCT	1	
J2	BATT 1S	S2B-PH-SM4-TB(LF)(SN)	JST	1	
J3	HANDLE	SM10B-GHS-TB(LF)(SN)	JST	1	
J4	MAZE	S4B-PH-SM4-TB(LF)(SN)	JST	1	
J5	OLED	PH1-04-UA	Adam Tech	1	
J6	QWIIC	SM04B-SRSS-TB(LF)(SN)	JST	1	
J7	microSD	DM3AT-SF-PEJM5	Hirose	1	
L1	4.7uH	XFL4020-472MEC	Coilcraft	1	
Q1,Q2	AO3401A	AO3401A	Alpha & Omega	2	
Q3,Q4	AO3400A	AO3400A	Alpha & Omega	2	
R1,R2	5.1k	RC0603FR-075R1kL	Yageo	2	
R3	1.5	RC0603FR-071R5L	Yageo	1	
R4	2.49k	RC0603FR-072K49L	Yageo	1	
R5	1	RC0603FR-071RL	Yageo	1	
R6	1k	RC0603FR-071kL	Yageo	1	
R7,R47,R48	2.2k	RC0603FR-072R2kL	Yageo	3	
R8,R9,R16,R33,R50,R59	100k	RC0603FR-07100kL	Yageo	6	DNP
R10,R11,R23,R24,R35,R38,R41,R44	1M	RC0603FR-071ML	Yageo	8	
R12	221k	RC0603FR-07221kL	Yageo	1	
R13	6.49k	RC0603FR-076R49kL	Yageo	1	
R14	49.9k	RC0603FR-0749R9kL	Yageo	1	
R15	0	RC0603JR-070RL	Yageo	1	
R17,R18,R19,R27,R28,R29,R30,R31,R32,R36,R37,R39,R40,R42,R43,R45,R46,R51,R53,R55,R60	10k	RC0603FR-0710kL	Yageo	21	
R20,R21,R22	470	RC0603FR-07470L	Yageo	3	
R25,R26,R34	4.7k	RC0603FR-074R7kL	Yageo	3	
R49,R58	100	RC0603FR-07100L	Yageo	2	

R52,R54,R56	360k	RC0603FR-07360kL	Yageo	3
R57	22	RC0805FR-0722RL	Yageo	1
SW1	POWER	PCM12SMTR	C&K	1
SW2	RESET	TL3342F160QG	E-Switch	1
SW3	BOOT/MODE	TL3342F160QG	E-Switch	1
SW4	START	TL3342F160QG	E-Switch	1
U1	USBLC6-2SC6	USBLC6-2SC6	STMicroelectronics	1
U3	LTC4054ES5-4.2	LTC4054ES5-4.2#TRPBF	Analog Devices	1
U4	LTC3536EMSE	LTC3536EMSE#PBF	Analog Devices	1
U5	ESP32-S3-WROOM-1-N8R2	ESP32-S3-WROOM-1-N8R2	Espressif	1
U6	MCP6004-I/SL	MCP6004T-I/SL	Microchip	1
U7	MCP3204-C/SL	MCP3204-CI/SL	Microchip	1
U8,U10	TPD4E05U06	TPD4E05U06DQAR	Texas Instruments	2
U9	PRTR5V0U2X	PRTR5V0U2X,215	Nexperia	1
U11	74LVC3G17	SN74LVC3G17DCUR	Texas Instruments	1
U12	ISM330DHCX	ISM330DHCXTR	STMicroelectronics	1